

Regex vs PCRE2 JIT — Benchmark Results

Ratio = PCRE2 JIT ÷ Regex. >1x = Regex faster. JIT compile time excluded.

OS	macOS-26.6.2-arm64-arm-64bit-Mach-O
Python	3.14.7
CPU	Apple M3 Pro
RAM	36 GB
Mojo	Mojo 1.1.0 (8189361e)

Benchmark	Regex (µs)	PCRE2 JIT (µs)	Ratio
Throughput scaling			
throughput_literal_100B	0.005	0.009	1.7x
throughput_literal_10KB	0.098	0.230	2.3x
throughput_literal_100KB	0.958	2.222	2.3x
throughput_literal_1MB	10.047	22.130	2.2x
throughput_class_10KB	0.304	2.504	8.2x
throughput_nomatch_100KB	0.968	6.968	7.2x
Anchors			
anchor_bol	0.002	0.007	4.4x
anchor_eol	0.006	0.009	1.6x
anchor_word_boundary	0.005	0.010	2.0x
anchor_word_boundary_miss	0.005	0.009	1.7x
anchor_bol_miss_10KB	0.001	0.007	8.8x
Multiline / DOTALL			
dotall_multiline_body	0.002	0.012	5.0x
Named groups			
named_group_date	0.004	0.014	3.4x
named_group_email	0.007	0.017	2.3x
positional_group_email	0.012	0.017	1.5x
Negative lookahead			
neg_lookahead	0.004	0.009	2.0x
neg_lookbehind	0.004	0.008	2.0x
password_validation_lookahead	0.019	0.026	1.4x
Alternation			
alternation_4	0.002	0.009	5.0x
alternation_4_search_2KB	0.191	2.014	10.6x
alternation_16	0.001	0.009	6.4x
alternation_16_miss	0.001	0.021	21.0x
Replace			
replace_50_matches	0.988	1.649	1.7x
replace_named_backref	0.040	0.083	2.1x

Benchmark	Regex (µs)	PCRE2 JIT (µs)	Ratio
Pathological patterns			
pathological_optional_16	0.011	94.930	8475.9x
pathological_dotstar_anchored_5K	0.120	1.980	16.4x
pathological_dotstar_miss_5K	0.001	0.110	100.0x
pathological_triple_backref	0.008	0.014	1.8x
pathological_nested_quantifier_miss	6.153	893.410	145.2x
Other			
static_email_search_2KB	0.071	1.287	18.1x
Real-world patterns			
realworld_url_parse	0.017	0.028	1.6x
realworld_phone	0.005	0.011	2.0x
realworld_hex_color	0.003	0.009	2.9x
realworld_semver	0.013	0.020	1.5x
realworld_ws_normalize	0.101	0.148	1.5x
realworld_log_search_1000_lines	0.346	0.713	2.1x
Inline flags			
inline_ignorecase	0.004	0.009	2.0x
inline_multiline_search	0.006	0.011	1.8x
Engine comparison			
engine_dfa_no_capture	0.004	0.010	2.4x
engine_pike_with_capture	0.006	0.015	2.6x
engine_backtrack_with_backref	0.007	0.011	1.6x
Regex faster: 41 slower: 0			